

```
tasks:
- name: Download dl-docker
  git:
    repo: https://github.com/saiprashanth/saiprashanth/dl-docker.git
    dest: "{{ DL_BUILD_DIR }}"

- name: Build image and with buildargs
  docker_image:
    path: "{{ DL_BUILD_DIR }}"
    name: "{{ DL_DOCKER_NAME }}"
    dockerfile: Dockerfile.cpu
    buildargs:
      tag: "{{ DL_DOCKER_NAME }}:cpu"
```

We first clone the deep learning Docker project sources. The `docker_image` module in Ansible helps us to build, load and pull images. We then use the `Dockerfile.cpu` file to build a Docker image targeting the CPU. If you have a GPU in your system, you can use the `Dockerfile.gpu` file. The above playbook can be invoked using the following command:

```
$ ansible-playbook playbooks/configuration/docker.yml -K
--tags=deep-learning
```

Depending on the CPU and RAM you have, it will take a considerable amount of time to build the image with all the software. So be patient!

Jupyter Notebook

The built `dl-docker` image contains Jupyter Notebook, which can be launched when you start the container. An Ansible playbook for the same is provided below:

```
- name: Start Jupyter notebook
  hosts: localhost
  gather_facts: true
  become: true
  tags: [notebook]

vars:
  DL_DOCKER_NAME: "floydhub/dl-docker"

tasks:
- name: Run container for Jupyter notebook
```

```
docker_container:
  name: "dl-docker-notebook"
  image: "{{ DL_DOCKER_NAME }}:cpu"
  state: started
  command: sh run_jupyter.sh
```

You can invoke the playbook using the following command:

```
$ ansible-playbook playbooks/configuration/docker.yml -K
--tags=notebook
```

The Dockerfile already exposes the port 8888, and hence you do not need to specify the same in the above `docker_container` configuration. After you run the playbook, using the `'docker ps'` command on the host system, you can obtain the container ID as indicated below:

```
$ sudo docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS
PORTS         NAMES
a876ad5af751   floydhub/dl-docker:cpu   "sh run_jupyter.sh"    11 minutes ago   Up 4 minutes        6006/tcp, 8888/tcp
tcp           dl-docker-notebook
```

You can now log in to the running container using the following command:

```
$ sudo docker exec -it a876 /bin/bash
```

You can then run an `'ifconfig'` command to find the local IP address ('172.17.0.2' in this case), and then open `http://172.17.0.2:8888` in a browser on your host system to see the Jupyter Notebook. A screenshot is shown in Figure 1.



Figure 1: Jupyter Notebook

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